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# COMMISSION STAFF WORKING DOCUMENT

# IMPACT ASSESSMENT REPORT

## ANNEX VIII-b

***Accompanying the***

**proposal for a Regulation of the European Parliament and of the Council**

**on nature restoration**

{COM(2022) 304 final} - {SEC(2022) 256 final} - {SWD(2022) 168 final}

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# Annex VIII-b: FORESTS

## Introduction

This paper provides information derived from the Member States' reports and assessments under Article 17 of the Habitats Directive. It is a background information to help identify possible restoration targets for the 'legal binding instrument' under the EU Biodiversity Strategy to 2030.

The 'forests' group includes, 69 Annex I habitat types (see Table 1): all forests with codes 9xxx (except wet/alluvial/riparian forests and wooded meadows, which were included in other groups).

The Habitats Directive defines Annex I forest habitats as: *(Sub)natural woodland vegetation comprising native species forming forests of tall trees, with typical undergrowth, and meeting the following criteria: rare or residual, and / or hosting species of Community interest*. The Interpretation Manual of EU Habitats[1] list the following additional criteria:

- *forests of native species;*
- *forests with a high degree of naturalness;*
- *forests of tall trees and high forest;*
- *presence of old and dead trees;*
- *forests with a substantial area;*
- *forests having benefited from continuous sustainable management over a significant period.*

Table 1 – Forest Annex I habitat types selected

| **Boreal forests (6 types)** | | **Temperate forests (cont.)** | |
|---|---|---|---|
| 9010 | Western Taïga | 91AA | Eastern white oak woods |
| 9020 | Fennoscandian hemiboreal natural old broad-leaved deciduous forests (*Quercus*, *Tilia*, *Acer*, *Fraxinus* or *Ulmus*) rich in epiphytes | 91BA | Moesian silver fir forests |
| 9030 | Natural forests of primary succession stages of landupheaval coast | 91CA | Rhodopide and Balkan Range Scots pine forests |
| 9040 | Nordic subalpine/subarctic forests with *Betula pubescens* ssp. *czerepanovii* | **Mediterranean and Macaronesian forests (18 types)** | |
| 9050 | Fennoscandian herb-rich forests with *Picea abies* | 9210 | Apeninne beech forests with *Taxus* and *Ilex* |
| 9060 | Coniferous forests on, or connected to, glaciofluvial eskers | 9220 | Apennine beech forests with *Abies alba* and beech forests with *Abies nebrodensis* |
| **Temperate forests (32 types)** | | 9230 | Galicio-Portuguese oak woods with *Quercus robur* and *Quercus pyrenaica* |

[1] https://ec.europa.eu/environment/nature/legislation/habitatsdirective/docs/Int_Manual_EU28.pdf

| | | | |
|---|---|---|---|
| 9110 | *Luzulo-Fagetum* beech forests | 9240 | *Quercus faginea* and *Quercus canariensis* Iberian woods |
| 9120 | Atlantic acidophilous beech forests with *Ilex* and sometimes also *Taxus* in the shrublayer (*Quercion robori-petraeae* or *Ilici-Fagenion*) | 9250 | *Quercus trojana* woods |
| 9130 | *Asperulo-Fagetum* beech forests | 9260 | *Castanea sativa* woods |
| 9140 | Medio-European subalpine beech woods with *Acer* and *Rumex arifolius* | 9270 | Hellenic beech forests with *Abies borisii-regis* |
| 9150 | Medio-European limestone beech forests of the *Cephalanthero-Fagion* | 9280 | *Quercus frainetto* woods |
| 9170 | *Galio-Carpinetum* oak-hornbeam forests | 9290 | Cupressus forests (*Acero-Cupression*) |
| 9180 | *Tilio-Acerion* forests of slopes, screes and ravines | 9310 | Aegean *Quercus brachyphylla* woods |
| 9190 | Old acidophilous oak woods with *Quercus robur* on sandy plains | 9320 | *Olea* and *Ceratonia* forests |
| 91A0 | Old sessile oak woods with *Ilex* and *Blechnum* in the British Isles | 9330 | *Quercus suber* forests |
| 91B0 | Thermophilous *Fraxinus angustifolia* woods | 9340 | *Quercus ilex* and *Quercus rotundifolia* forests |
| 91C0 | Caledonian forest | 9350 | *Quercus macrolepis* forests |
| 91G0 | Pannonic woods with *Quercus petraea* and *Carpinus betulus* | 9360 | Macaronesian laurel forests (*Laurus*, *Ocotea*) |
| 91H0 | Pannonian woods with *Quercus pubescens* | 9380 | Forests of *Ilex aquifolium* |
| 91I0 | Euro-Siberian steppic woods with *Quercus* spp. | 9390 | Scrub and low forest vegetation with *Quercus alnifolia* |
| 91J0 | *Taxus baccata* woods of the British Isles | 93A0 | Woodlands with *Quercus infectoria* (*Anagyro foetidae-Quercetum infectoriae*) |
| 91K0 | Illyrian *Fagus sylvatica* forests (*Aremonio-Fagion*) | **Mountainous coniferous forests (13 types)** | |
| 91L0 | Illyrian oak-hornbeam forests (*Erythronio-Carpinion*) | 9410 | Acidophilous *Picea* forests of the montane to alpine levels (*Vaccinio-Piceetea*) |
| 91M0 | Pannonian-Balkanic turkey oak –sessile oak forests | 9420 | Alpine *Larix decidua* and/or *Pinus cembra* forests |
| 91P0 | Holy Cross fir forest (*Abietetum polonicum*) | 9430 | Subalpine and montane *Pinus uncinata* forests (* if on gypsum or limestone) |
| 91Q0 | Western Carpathian calcicolous Pinus sylvestris forests | 9510 | Southern Apennine Abies alba forests |
| 91R0 | Dinaric dolomite Scots pine forests (*Genisto januensis-Pinetum*) | 9520 | *Abies pinsapo* forests |
| 91S0 | Western Pontic beech forests | 9530 | (Sub-) Mediterranean pine forests with endemic black pines |
| 91T0 | Central European lichen Scots pine forests | 9540 | Mediterranean pine forests with endemic Mesogean pines |
| 91U0 | Sarmatic steppe pine forest | 9550 | Canarian endemic pine forests |
| 91V0 | Dacian Beech forests (*Symphyto-Fagion*) | 9560 | Endemic forests with *Juniperus* spp. |
| 91W0 | Moesian beech forests | 9570 | *Tetraclinis articulata* forests |

| | | | |
|---|---|---|---|
| 91X0 | Dobrogean beech forests | 9580 | Mediterranean *Taxus baccata* woods |
| 91Y0 | Dacian oak & hornbeam forests | 9590 | Cedrus brevifolia forests (*Cedrosetum brevifoliae*) |
| 91Z0 | Moesian silver lime woods | 95A0 | High oro-Mediterranean pine forests |

## Forest habitats coverage in the EU

The 69 habitat types selected cover close to **357 952 km²** (**9.2 %** of the EU terrestrial area[2]); this excludes areas reported by Romania, which are known to be largely overestimated[3].

The data available from Corine Land Cover[4] and from the Ecosystems Map of Europe[5] do not allow a straightforward comparison between the total area of forests in the EU and the area covered by Annex I forests. This is mainly due to the nomenclatures used and the spatial resolution of the datasets. A comparison between these data sources – excluding alluvial and riparian forests – is given in Table 2 below.

### Table 2 – Forest areas (km²) from different sources (EU27)

Corine Land Cover 2018 (level 3)

| | |
|---|---|
| **Forests** | **1 356 423** |
| 311 – Broad-leaved forest | 435 974 |
| 312 – Coniferous forest | 660 642 |
| 313 – Mixed forest | 259 807 |

Ecosystems map (level 3)

| | |
|---|---|
| **G – Woodland, forest and other wooded land** | **1 770 997** |
| G1 – Broad-leaved deciduous woodland | 682 357 |
| G2 – Broad-leaved evergreen woodland | 52 200 |
| G3 – Coniferous woodland | 707 302 |
| G4 – Mixed deciduous and coniferous woodland | 236 096 |
| G5 – Lines of trees, small anthropogenic woodlands, recently felled woodland, early-stage woodland and coppice | 93 042 |

The areas of Annex I forests, have a good representation in most EU countries (except Ireland, Malta and the Netherlands) (see Map 1). The Member States with the highest proportion of those habitats are Slovenia (33 %), Croatia (24 %), Bulgaria (23 %), Greece (22 %), Cyprus (16 %), Austria (15 %), Italy (14 %), Slovakia (14 %), Spain (11 %), France (11 %) and Luxembourg (10 %); seven Member States have less than 2 % of their territory covered by forest Annex I habitats.

[2] Area of habitats calculated from the area reported by Member States as 'best estimate' or 'average of minimum/maximum'

[3] The average total area of forest habitats reported by Romania is 59 126 km²

[4] https://www.eea.europa.eu/data-and-maps/dashboards/land-cover-and-change-statistics

[5] https://www.eea.europa.eu/themes/biodiversity/mapping-europes-ecosystems



Source: European Environment Agency – April 2021

Table 3 gives the areas and proportion of Annex I forests for each Member State, including coverage by Natura 2000. Maps illustrating the distribution of different forest habitats in the EU are available in Annex A.

From the 357 952 $km^2$ of forest habitats (excluding Romania), **38 %** is estimated to be inside the Natura 2000 network (about **135 596 $km^2$**); this may be an underestimation since reports from Member States were not comprehensive on this regard. The coverage by Natura 2000 varies according to the sub-group, **from 52 %** for 'boreal forests' **to 31 %** for 'mountainous forests'. The proportion of habitats per sub-group of forest and their coverage is detailed in Table 4.

Coverage by Natura 2000 also greatly varies according to the Member State: **from near 85 %** (Estonia) **to about 9 %** (France) (Table 3).

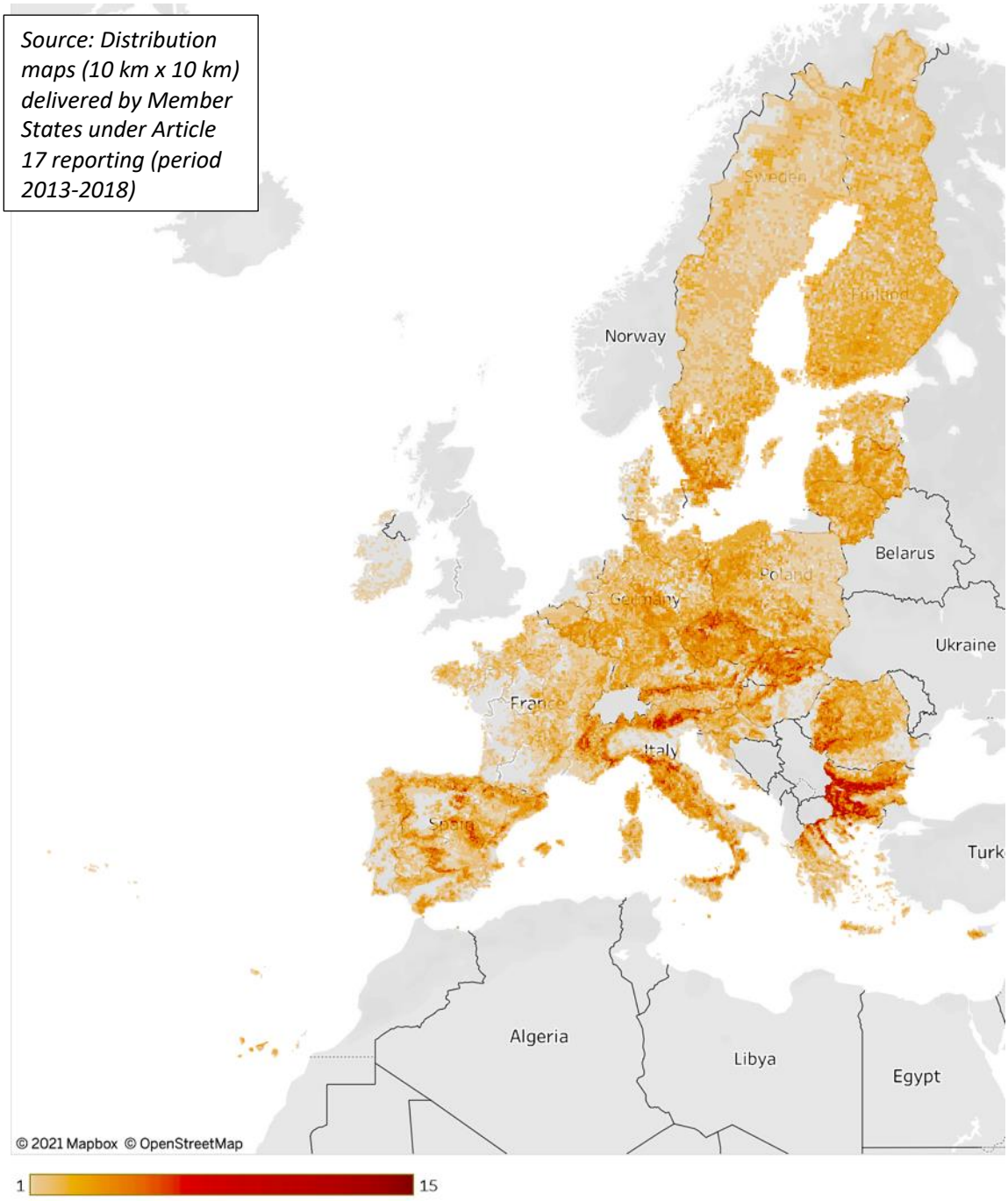
However, several Member States reported over 75 % of Annex I forest habitats area inside Natura 2000 (Croatia, Estonia, Malta and Poland).



Source: European Environment Agency – April 2021

## Map 1 – Distribution of the 69 Annex I forest habitats in the EU

*Source: Distribution maps (10 km x 10 km) delivered by Member States under Article 17 reporting (period 2013-2018)*




Note: the shades of brown indicate the number of habitat types per 10 km x 10 km grid cell.



Source: European Environment Agency – April 2021

**Table 3 – Area and proportion of forest habitats per Member State**

| | Member State area ($km^2$) | In the Member State | | Proportion of the forest habitats area (%) | Inside Natura 2000 | |
|---|---|---|---|---|---|---|
| | | Forest habitats area ($km^2$) | Forest habitats area (%) | | Forest habitats area | % Forest habitats area |
| Austria | 83 944 | 12 838.2 | 15.3 | 3.6 | 2 170.6 | 16.9 |
| Belgium | 30 683 | 1 409.1 | 4.6 | 0.4 | 757.6 | 53.8 |
| Bulgaria | 110 995 | 25 039.4 | 22.6 | 7.0 | 13 795.4 | 55.1 |
| Croatia | 55 590 | 13 278.9 | 23.9 | 3.7 | 10 025.1 | 75.5 |
| Cyprus | 9 249 | 1 501.3 | 16.2 | 0.4 | 545.2 | 36.3 |
| Czechia | 78 874 | 4 570.5 | 5.8 | 1.3 | 2 016.0 | 44.1 |
| Denmark | 44 162 | 643.5 | 1.5 | 0.2 | 117.2 | 18.2 |
| Estonia | 45 382 | 897.0 | 2.0 | 0.3 | 758.5 | 84.6 |
| Finland | 338 004 | 27 337.9 | 8.1 | 7.6 | 13 762.7 | 50.3 |
| France | 551 881 | 58 711.1 | 10.6 | 16.4 | 5 383.2 | 9.2 |
| Germany | 362 177 | 16 175.8 | 4.5 | 4.5 | 6 305.5 | 39.0 |
| Greece | 132 014 | 28 791.0 | 21.8 | 8.0 | 7 650.0 | 26.6 |
| Hungary | 93 012 | 4 003.0 | 4.3 | 1.1 | 2 679.0 | 66.9 |
| Ireland | 70 699 | 60.9 | 0.1 | 0.0 | 40.2 | 66.0 |
| Italy | 301 321 | 42 405.3 | 14.1 | 11.8 | 13 408.2 | 31.6 |
| Latvia | 64 590 | 968.6 | 1.5 | 0.3 | 309.2 | 31.9 |
| Lithuania | 65 289 | 1 204.1 | 1.8 | 0.3 | 399.0 | 33.1 |
| Luxembourg | 2 595 | 260.9 | 10.1 | 0.1 | 129.5 | 49.6 |
| Malta | 316 | 0.7 | 0.2 | 0.0 | 0.5 | 80.8 |
| Netherlands | 39 898 | 156.9 | 0.4 | 0.0 | 99.2 | 63.2 |
| Poland | 312 683 | 8 070.0 | 2.6 | 2.3 | 6 378.3 | 79.0 |
| Portugal | 92 378 | 715.5 | 0.8 | 0.2 | 515.0 | 72.0 |
| *Romania (*)* | *238 404* | 59 126.3 | 24.8 | | 20 017.6 | 33.9 |
| Slovakia | 49 026 | 6 726.6 | 13.7 | 1.9 | 2 210.3 | 32.9 |
| Slovenia | 20 274 | 6 701.5 | 33.1 | 1.9 | 3 396.4 | 50.7 |
| Spain | 506 222 | 57 153.3 | 11.3 | 16.0 | 22 610.0 | 39.6 |
| Sweden | 450 110 | 38 331.4 | 8.5 | 10.7 | 20 134.2 | 52.5 |
| Total | 4 149 772 | 417 078.7 | 10.1 | | 155 613.8 | 37.3 |
| **Total (without Romania)** | **3 911 772** | **357 952.4** | **9.2** | | **135 596.2** | **37.9** |
| | | | | | | |

Notes: Member States with more than 9.2 % (the EU average) of their terrestrial area covered by wetlands are highlighted; (*) areas reported by Romania are overestimated.



Source: European Environment Agency – April 2021

**Table 4 – Area and proportion of forest habitats per sub-group**

| | | Inside Natura 2000 | |
|---|---|---|---|
| **EU27 excluding Romania** | **Area (km²)** | **Forest area (km²)** | **% forest area** |
| **Boreal forests** | **68 286** | **35 184** | **52** |
| 9010 | 36 315 | 22 199 | 61 |
| 9020 | 440 | 196 | 45 |
| 9030 | 350 | 139 | 40 |
| 9040 | 19 600 | 11 622 | 59 |
| 9050 | 4 464 | 560 | 13 |
| 9060 | 7 116 | 468 | 7 |
| **Temperate forests** | **172 384** | **59 752** | **35** |
| 9110 | 21 009 | 6 113 | 29 |
| 9120 | 15 384 | 2 530 | 16 |
| 9130 | 56 043 | 13 692 | 24 |
| 9140 | 612 | 161 | 26 |
| 9150 | 5 845 | 2 613 | 45 |
| 9170 | 8 347 | 4 320 | 52 |
| 9180 | 2 472 | 1 095 | 44 |
| 9190 | 1 785 | 879 | 49 |
| 91A0 | 61 | 40 | 65 |
| 91AA | 6 007 | 1 193 | 20 |
| 91B0 | 333 | 191 | 57 |
| 91BA | 245 | 188 | 77 |
| 91CA | 2 440 | 1 432 | 59 |
| 91G0 | 3 239 | 1 760 | 54 |
| 91H0 | 938 | 550 | 59 |
| 91I0 | 1 515 | 1 200 | 79 |
| 91J0 | 1 | 1 | 100 |
| 91K0 | 16 338 | 11 475 | 70 |
| 91L0 | 3 427 | 1 251 | 37 |
| 91M0 | 22 986 | 7 185 | 31 |
| 91P0 | 185 | 157 | 85 |
| 91Q0 | 20 | 13 | 63 |
| 91R0 | 85 | 48 | 57 |
| 91S0 | 268 | 222 | 83 |
| 91T0 | 277 | 99 | 36 |
| 91U0 | 9 | 4 | 49 |
| 91W0 | 2 124 | 1 143 | 54 |
| 91Z0 | 389 | 197 | 51 |
| **Mediterranean & Macaronesian forests** | **66 335** | **24 911** | **38** |
| 9210 | 3 271 | 1 916 | 59 |
| 9220 | 347 | 280 | 81 |
| 9230 | 13 154 | 3 170 | 24 |
| 9240 | 3 445 | 1 544 | 45 |
| 9250 | 457 | 452 | 99 |
| 9260 | 8 528 | 2 034 | 24 |
| 9270 | 570 | 570 | 100 |
| 9280 | 996 | 221 | 22 |



Source: European Environment Agency – April 2021

| | | Inside Natura 2000 | |
|---|---|---|---|
| **EU27 excluding Romania** | **Area ($km^2$)** | **Forest area ($km^2$)** | **% forest area** |
| 9290 | 438 | 175 | 40 |
| 9310 | 3 | 3 | 100 |
| 9320 | 1 966 | 1 228 | 62 |
| 9330 | 4 284 | 2 143 | 50 |
| 9340 | 27 784 | 10 545 | 38 |
| 9350 | 297 | 79 | 27 |
| 9360 | 603 | 398 | 66 |
| 9380 | 123 | 121 | 99 |
| 9390 | 61 | 31 | 50 |
| 93A0 | 6 | 2 | 34 |
| **Mountainous forests** | **50 947** | **15 748** | **31** |
| 9410 | 12 376 | 3 828 | 31 |
| 9420 | 4 007 | 981 | 24 |
| 9430 | 1 516 | 901 | 59 |
| 9510 | 83 | 75 | 91 |
| 9520 | 13 | 13 | 100 |
| 9530 | 7 896 | 1 918 | 24 |
| 9540 | 21 677 | 5 506 | 25 |
| 9550 | 737 | 626 | 85 |
| 9560 | 2 153 | 1 571 | 73 |
| 9570 | 1 | 1 | 83 |
| 9580 | 16 | 17 | 110 |
| 9590 | 3 | 3 | 90 |
| 95A0 | 470 | 309 | **31** |
| **TOTAL** | **357 952** | **135 596** | **38** |



Source: European Environment Agency – April 2021

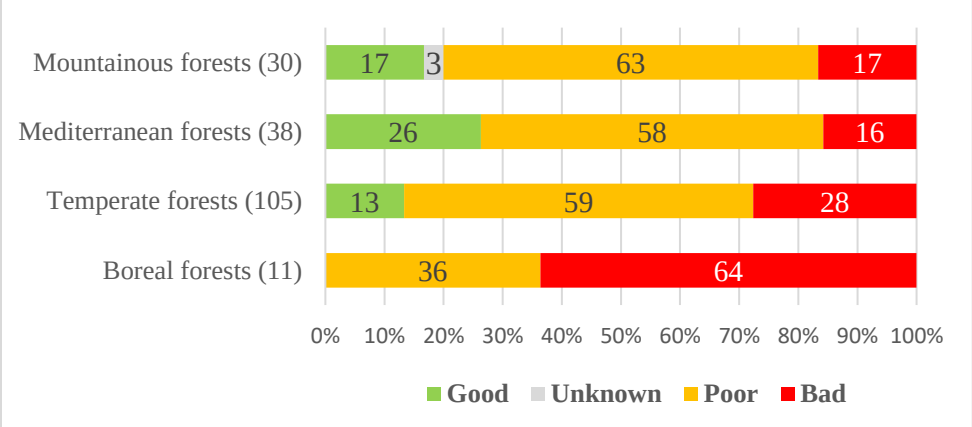
## Conservation status and trends

The vast majority (**84 %**) of the assessments of the 69 forest habitats at the EU level have an **unfavourable** conservation status (58 % poor and 26 % bad). Only **16 %** have a **good** conservation status. There are some differences between the different habitat groups (Figure 1): 'Mediterranean forests' has the highest proportion of good status (26 %) and the 'Boreal forests' the worst status (100 % unfavourable).

Among the habitat assessments that do not have a good status, under one-fifth have a **deteriorating** trend (**17 %**) while **18 %** have an improving trend. Near half have a stable trend – not improving nor deteriorating – (**47 %)**; the conservation status trend is unknow for **17 %** of the assessments. The group with the worst conservation status trends is '**Boreal forests**' (**45 % deteriorating**); however, '**Mountainous forests**' have the higher proportion of **improving** trends (**28** %) (Figure 2).

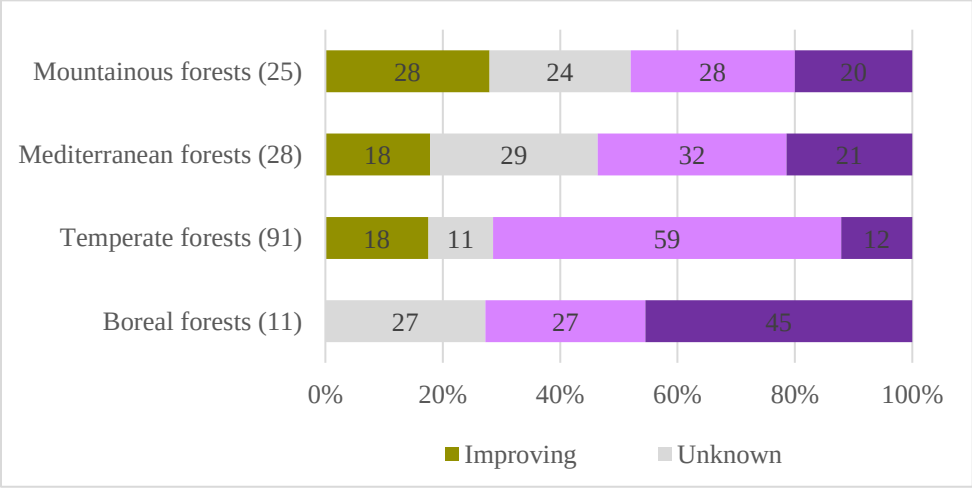
Details on conservation status and conservation status trends for each Member State are given in Table 5.

**Figure 1 – Conservation status of forests at the EU level (in percentage)**




Note: Number of assessments per group shown in brackets.

**Figure 2 – Conservation status trends of forests at the EU level (in percentage)**




Note: Number of assessments per group shown in brackets.



Source: European Environment Agency – April 2021

**Table 5 – Conservation status and trends of forest habitats in the Member States (in percentage)**

| Member States | FV | U1- | U1+ | U1= | U1x | U2- | U2+ | U2= | U2x | XX |
|---|---|---|---|---|---|---|---|---|---|---|
| AT (25) | 16 | 8 | 16 | 20 | 20 | | | 8 | 8 | 4 |
| BE (11) | | | 18 | 9 | 9 | | 18 | 27 | 18 | |
| BG (51) | 31 | | 55 | 8 | 4 | | | | 2 | |
| CY (8) | 63 | | | 13 | | | | 25 | | |
| CZ (18) | | 6 | | 44 | | 33 | | 6 | 6 | 6 |
| DE (26) | 46 | 12 | 8 | 8 | | 19 | | 4 | | 4 |
| DK (10) | | | | | | 70 | | | 30 | |
| EE (5) | 20 | 60 | | | | | | | 20 | |
| ES (44) | 7 | 2 | 7 | 34 | 27 | 7 | | 5 | 7 | 5 |
| FI (11) | 9 | 18 | | 45 | | 18 | | | 9 | |
| FR (47) | 23 | 4 | 4 | 32 | 9 | 6 | | 6 | 9 | 6 |
| GR (22) | 73 | | 18 | 5 | | | 5 | | | |
| HR (23) | 78 | | | 13 | | 9 | | | | |
| HU (10) | 10 | | | 50 | | 10 | | 30 | | |
| IE (2) | | | | | | 50 | | 50 | | |
| IT (58) | 10 | 19 | 2 | 33 | 9 | 16 | | 7 | 2 | 3 |
| LT (7) | 14 | | | | 29 | 29 | | | 29 | |
| LU (4) | 75 | | | 25 | | | | | | |
| LV (6) | | | | | 33 | | | 17 | 50 | |
| MT (4) | 25 | | | 25 | | | | 50 | | |
| NL (3) | | | 33 | 33 | | 33 | | | | |
| PL (19) | 32 | 5 | | 42 | | 11 | | 11 | | |
| PT (16) | 6 | 19 | | 25 | | 50 | | | | |
| RO (35) | 63 | | | 23 | | | | 6 | 3 | 6 |
| SE (18) | 6 | 6 | | 6 | 17 | 11 | | 28 | 28 | |
| SI (13) | 31 | 15 | | 54 | | | | | | |
| SK (23) | 43 | | | 35 | 9 | 9 | | | 4 | |

Notes: FV = good, U1 = poor, U2 = bad, XX = unknown conservation status
'-' = deteriorating, '+' = improving, '=' = stable, 'x' = unknown conservation status trend; number of assessments per Member State shown in brackets.



Source: European Environment Agency – April 2021

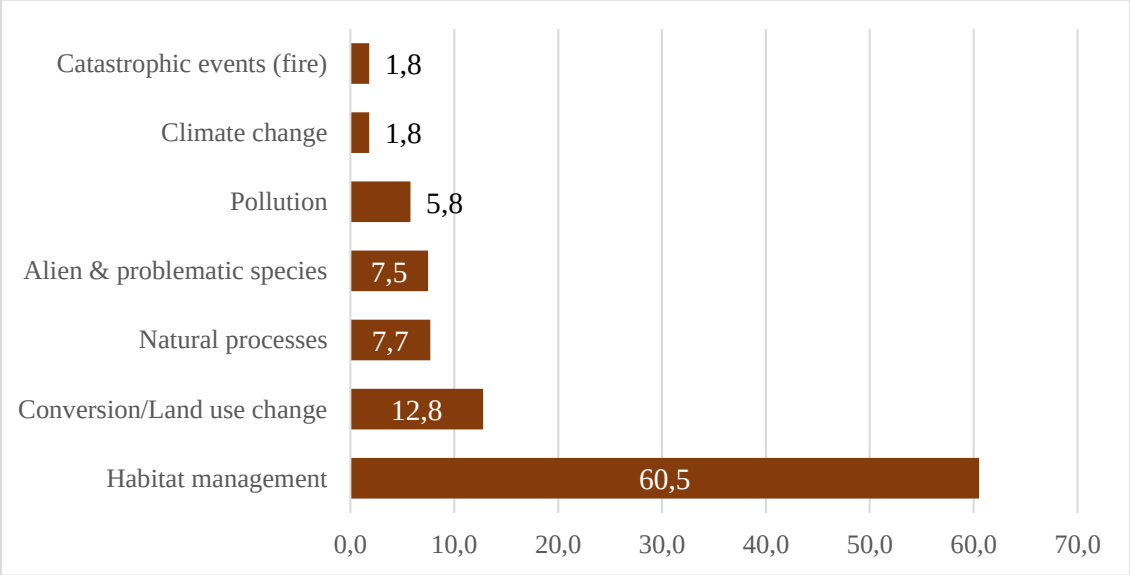
## Pressures

Forest habitats are subject to a wide diversity of pressures resulting in their degradation and extirpation. According to Member States reports under Article 17 of the Habitats Directive, the top three groups of pressures (in percentage of the total) are (Figure 3):

- **Habitat management** with close to **61 %** of all pressures; these include inadequate forestry practices like removal of dead and old trees (30 %), clear-cutting (10 %), reduction of old growth forest (8 %)
- **Conversion and land use change** amounts to **13 %**; from these, 45 % correspond to conversion to other forest types (including monocultures), 22 % to construction of urban, commercial, industrial and leisure areas, and 12 % to transport infrastructure
- **Natural processes**, with about **8 %**; this is mainly due to interspecific relations (competition, parasitism and pathogens) (43 %) and changes in species composition (34 %)

Equally important is **alien and problematic species** with over **7 %**, mainly invasive alien species (58 %), and plant diseases, pathogens, and pests (26 %).

**Figure 3 – Pressures reported for forest habitats (in percentage)**




Note: based on pressures reported as 'high-ranking'.



Source: European Environment Agency – April 2021

## Condition of habitats

Member States reported on the condition of habitat types under Article 17 of the Habitats Directive. This data can be used to estimate the area of forest habitats assessed as degraded (condition not-good) therefore, requiring restoration.

The area of Annex I forest habitats that would need to be restored, i.e., improved condition, is **at least 79 210 km$^2$**, representing **22 %** of the total area reported for this group of habitats (the values exclude Romania). However, the condition of habitats reported as 'unknown' (or not reported) is over 116 444 km$^2$ (33 % of the total area). This means that the area requiring restoration is bigger than 79 210 km$^2$; for example, assuming that half of the 'unknown' area is in a not-good condition, the area to be restored would be over 137 000 km$^2$ or 195 000 km$^2$ if all the 'unknown' is assumed to be in a 'not-good' condition (22 % of the total area). Table 6 gives information for each of the 69 forest habitats (excluding Romania) and Table 7 the condition areas and percentage for each of the Member States.

In addition to the habitat condition, Member States also reported on the 'favourable reference areas'[6]. Comparing these areas with the actual habitat areas allows to estimate how much area of the habitat would need to be re-created to achieve a good distribution and area of the habitat. Based on this data, it is estimated that a **strict minimum** of **3 500 km$^2$** would need to be **re-created** to achieve a 'favourable area'.

- 830 km$^2$ of Boreal forest habitats
- 1 050 km$^2$ of Temperate forest habitats
- 1 270 km$^2$ of Mediterranean forest habitats
- 350 km$^2$ of Mountainous forest habitats

However, these values are much higher since several Member States did not provide enough information in their reports to allow a more realistic estimation.

[6] The surface area in a given biogeographical region considered the minimum necessary to ensure the long-term viability of the habitat type; this should include necessary areas for restoration or development for those habitat types for which the present coverage is not sufficient to ensure long-term viability



Source: European Environment Agency – April 2021

## Table 6 – Condition of forest habitats per Annex I habitat type

| | | Condition (area in km2) | | | Condition (in percentage) | | |
|---|---|---|---|---|---|---|---|
| | **Habitat area** | **Good** | **Not-good** | **Unknown** | **Good** | **Not-good** | **Unknown** |
| **Total** | **357 952** | **162 296** | **79 211** | **116 444** | **45** | **22** | **33** |
| **Boreal forests** | **68 286** | **16 160** | **6 728** | **45 398** | **24** | **10** | **66** |
| 9010 | 36 315 | 13 774 | 3 568 | 18 973 | 38 | 10 | 52 |
| 9020 | 440 | 118 | 120 | 202 | 27 | 27 | 46 |
| 9030 | 350 | 121 | 147 | 82 | 35 | 42 | 23 |
| 9040 | 19 600 | 1 452 | 2 590 | 15 558 | 7 | 13 | 79 |
| 9050 | 4 464 | 646 | 249 | 3 569 | 14 | 6 | 80 |
| 9060 | 7 116 | 49 | 54 | 7 013 | 1 | 1 | 99 |
| **Temperate forests** | **172 384** | **92 798** | **51 610** | **27 976** | **54** | **30** | **16** |
| 9110 | 21 009 | 12 567 | 4 585 | 3 857 | 60 | 22 | 18 |
| 9120 | 15 384 | 2 294 | 11 883 | 1 207 | 15 | 77 | 8 |
| 9130 | 56 043 | 22 018 | 26 574 | 7 451 | 39 | 47 | 13 |
| 9140 | 612 | 148 | 51 | 414 | 24 | 8 | 68 |
| 9150 | 5 845 | 2 108 | 2 007 | 1 730 | 36 | 34 | 30 |
| 9170 | 8 347 | 5 662 | 2 044 | 641 | 68 | 24 | 8 |
| 9180 | 2 472 | 1 346 | 206 | 919 | 54 | 8 | 37 |
| 9190 | 1 785 | 727 | 1 042 | 16 | 41 | 58 | 1 |
| 91A0 | 61 | 43 | 18 | | 70 | 30 | |
| 91AA | 6 007 | 1 359 | 98 | 4 550 | 23 | 2 | 76 |
| 91B0 | 333 | 39 | 49 | 245 | 12 | 15 | 74 |
| 91BA | 245 | 246 | | | 100 | | |
| 91CA | 2 440 | 2 421 | | 18 | 99 | | 1 |
| 91G0 | 3 239 | 2 541 | 482 | 215 | 78 | 15 | 7 |
| 91H0 | 938 | 640 | 81 | 217 | 68 | 9 | 23 |
| 91I0 | 1 515 | 1 385 | 85 | 46 | 91 | 6 | 3 |
| 91J0 | 1 | | 1 | | 35 | 65 | |
| 91K0 | 16 338 | 13 553 | 1 282 | 1 503 | 83 | 8 | 9 |
| 91L0 | 3 427 | 2 577 | 285 | 565 | 75 | 8 | 16 |
| 91M0 | 22 986 | 18 035 | 664 | 4 288 | 78 | 3 | 19 |
| 91P0 | 185 | 123 | 62 | | 66 | 34 | |
| 91Q0 | 20 | 5 | 1 | 14 | 25 | 6 | 69 |
| 91R0 | 85 | 82 | 4 | | 96 | 4 | |
| 91S0 | 268 | 268 | | | 100 | | |
| 91T0 | 277 | 93 | 104 | 80 | 34 | 37 | 29 |
| 91U0 | 9 | 5 | 3 | 1 | 50 | 37 | 13 |
| 91W0 | 2 124 | 2 124 | | | 100 | | |
| 91Z0 | 389 | 389 | | | 100 | | |



Source: European Environment Agency – April 2021

| | | Condition (area in km2) | | | Condition (in percentage) | | |
|---|---|---|---|---|---|---|---|
| | **Habitat area** | **Good** | **Not-good** | **Unknown** | **Good** | **Not-good** | **Unknown** |
| **Mediterranean & Macaronesian forests** | **66 335** | **28 921** | **14 727** | **22 687** | **44** | **22** | **34** |
| 9210 | 3 271 | 1 858 | 40 | 1 373 | 57 | 1 | 42 |
| 9220 | 347 | 275 | 4 | 68 | 79 | 1 | 20 |
| 9230 (*) | 13 154 | 8 945 | 7 378 | -3 170 | 68 | 56 | -24 |
| 9240 | 3 445 | 388 | 508 | 2 549 | 11 | 15 | 74 |
| 9250 (#) | 457 | 572 | 1 | -115 | 125 | | -25 |
| 9260 | 8 528 | 1 956 | 511 | 6 060 | 23 | 6 | 71 |
| 9270 | 570 | 514 | | 56 | 90 | | 10 |
| 9280 | 996 | 896 | | 100 | 90 | | 10 |
| 9290 | 438 | 394 | 1 | 43 | 90 | | 10 |
| 9310 | 3 | | 1 | 3 | | 20 | 80 |
| 9320 | 1 966 | 1 280 | 230 | 456 | 65 | 12 | 23 |
| 9330 | 4 284 | 1 540 | 1 470 | 1 273 | 36 | 34 | 30 |
| 9340 | 27 784 | 9 600 | 4 395 | 13 789 | 35 | 16 | 50 |
| 9350 | 297 | 267 | | 30 | 90 | | 10 |
| 9360 | 603 | 312 | 156 | 135 | 52 | 26 | 22 |
| 9380 | 123 | 65 | 23 | 35 | 53 | 18 | 28 |
| 9390 | 61 | 55 | 6 | | 90 | 10 | |
| 93A0 | 6 | 3 | 3 | | 46 | 54 | |
| **Mountainous coniferous forests** | **50 947** | **24 418** | **6 146** | **20 383** | **48** | **12** | **40** |
| 9410 | 12 376 | 8 383 | 324 | 3 670 | 68 | 3 | 30 |
| 9420 | 4 007 | 1 118 | 102 | 2 786 | 28 | 3 | 70 |
| 9430 | 1 516 | 228 | 1 087 | 201 | 15 | 72 | 13 |
| 9510 | 83 | 74 | 2 | 8 | 89 | 2 | 9 |
| 9520 | 13 | 8 | 3 | 2 | 59 | 24 | 16 |
| 9530 | 7 896 | 3 315 | 739 | 3 842 | 42 | 9 | 49 |
| 9540 | 21 677 | 9 806 | 3 524 | 8 346 | 45 | 16 | 39 |
| 9550 | 737 | 626 | | 111 | 85 | | 15 |
| 9560 | 2 153 | 661 | 278 | 1 214 | 31 | 13 | 56 |
| 9570 | 1 | 1 | | | 68 | 19 | 13 |
| 9580 | 16 | 11 | 1 | 4 | 71 | 4 | 25 |
| 9590 | 3 | 3 | | | 100 | | |
| 95A0 | 470 | 185 | 86 | 199 | 39 | 18 | 42 |

Notes: Areas reported by Romania excluded from the table; (*) issue with data for habitat 9230 in the Atlantic region of France (areas repeated in 'good' and +not-good' condition); (#) (*) issue with data for habitat 9250 in the Mediterranean region of Italy (area in 'good' condition bigger than total area of the habitat).



Source: European Environment Agency – April 2021

**Table 7 – Condition of Annex I forest habitats per Member State**

| | Habitats' area (km$^2$) | | | | Percentage | | |
|---|---|---|---|---|---|---|---|
| **Member State** | **Total** | **Good** | **Not-good** | **Unknown** | **Good** | **Not-good** | **Unknown** |
| AT | 12 838 | 11 612 | 590 | 636 | 90 | 5 | 5 |
| BE | 1 409 | 53 | 192 | 1 164 | 4 | 14 | 83 |
| BG | 25 039 | 25 038 | 1 | | 100 | | |
| CY | 1 501 | 1 285 | 216 | | 86 | 14 | |
| CZ | 4 570 | 3 235 | 461 | 874 | 71 | 10 | 19 |
| DE | 16 176 | 14 517 | 1 129 | 530 | 90 | 7 | 3 |
| DK | 644 | 2 | 642 | | | 100 | |
| EE | 897 | 586 | 290 | 21 | 65 | 32 | 2 |
| ES | 57 153 | 13 314 | 13 608 | 30 231 | 23 | 24 | 53 |
| FI | 27 338 | 11 804 | 4 901 | 10 633 | 43 | 18 | 39 |
| FR (#) | 58 711 | 15 968 | 46 922 | -4 178 | 27 | 80 | -7 |
| GR | 28 791 | 25 122 | 204 | 3 465 | 87 | 1 | 12 |
| HR | 13 279 | 13 265 | 14 | | 100 | | |
| HU | 4 003 | 1 894 | 1 520 | 590 | 47 | 38 | 15 |
| IE | 61 | 42 | 19 | | 69 | 31 | |
| IT | 42 405 | 10 783 | 845 | 30 778 | 25 | 2 | 73 |
| LT | 1 204 | | | 1 204 | | | 100 |
| LU | 261 | 241 | 1 | 19 | 92 | | 7 |
| LV | 969 | 613 | 355 | | 63 | 37 | |
| MT | 1 | | | | 70 | 30 | |
| NL | 157 | 76 | 81 | | 49 | 51 | |
| PL | 8 070 | 3 936 | 4 133 | 1 | 49 | 51 | |
| PT | 716 | 297 | 261 | 158 | 41 | 36 | 22 |
| *RO (*)* | 59 126 | 55 410 | 3 401 | 315 | 94 | 6 | 1 |
| SE | 38 331 | 3 321 | 1 250 | 33 760 | 9 | 3 | 88 |
| SI | 6 702 | 5 181 | 1 508 | 13 | 77 | 22 | |
| SK | 6 726 | 113 | 69 | 6 544 | 2 | 1 | 97 |

Notes: (*) areas reported by Romania largely overestimated; (#) issue with data for several habitats (repeated areas for 'good' and 'not-good condition' – 9230, 9330)



Source: European Environment Agency – April 2021

## Carbon stock and sequestration

Forests are the dominant ecosystem in terms of area, sequestration and carbon storage potential. Annual sequestration of forest habitats reaches 100 Mio tons C per year equivalent to 367 Mio tons of $CO_2$. The storage capacity is estimated to range between 2,8 and 8,8 Gt of carbon equivalent to 10,3 Gt and 32,3 Gt of $CO_2$. Despite area covered by forest is more than twice as much as wetland area and sequestration rate is five times higher, the storage capacity for forests per $km^2$ is slightly lower compared to wetlands. Forests acumulate carbon over decades to centuries before reaching saturation point. If harvested, the climate change mitigation effect is determinated by the use of wood and timber (e.g. short-term storage if used for fuel or papel versus long-term storage in furniture and construction). Harvesting also leads to loss of soil organic carbon by erosion and mineralization.



Source: European Environment Agency – April 2021

## Table 8 – Carbon stock and sequestration of Annex I forest habitats

| **EU27 excluding Romania** | Forest area ($km^2$) | Total Carbon Stock (Mt) | | Potential carbon sequestration rate (Mt $y^{-1}$) |
|---|---|---|---|---|
| | EU-27 | min | max | mean |
| **Boreal forests** | **68 285.55** | **512.14** | **1536.42** | **20.52** |
| 9010 | 36 315.02 | 272.36 | 817.09 | 10.89 |
| 9020 | 440.22 | 3.30 | 9.90 | 0.17 |
| 9030 | 350.00 | 2.63 | 7.88 | 0.11 |
| 9040 | 19 600.00 | 147.00 | 441.00 | 5.88 |
| 9050 | 4 464.06 | 33.48 | 100.44 | 1.34 |
| 9060 | 7 116.26 | 53.37 | 160.12 | 2.13 |
| **Temperate forests** | **172 384.14** | **1293.96** | **4756.65** | **56.88** |
| 9110 | 21 009.04 | 157.57 | 630.27 | 7.88 |
| 9120 | 15 383.62 | 115.38 | 461.51 | 5.77 |
| 9130 | 56 042.97 | 420.32 | 1681.29 | 21.02 |
| 9140 | 612.33 | 4.59 | 18.37 | 0.23 |
| 9150 | 5 844.65 | 43.83 | 175.34 | 2.19 |
| 9170 | 8 347.21 | 62.60 | 187.81 | 3.13 |
| 9180 | 2 471.91 | 18.54 | 55.62 | 0.93 |
| 9190 | 1 785.03 | 13.39 | 53.55 | 0.54 |
| 91A0 | 61.08 | 0.46 | 1.83 | 0.02 |
| 91AA | 6 006.88 | 45.05 | 135.15 | 0.90 |
| 91B0 | 333.10 | 2.50 | 7.49 | 0.05 |
| 91BA | 245.29 | 3.68 | 7.36 | 0.11 |
| 91CA | 2 439.55 | 18.30 | 36.59 | 0.55 |
| 91G0 | 3 238.73 | 24.29 | 72.87 | 1.21 |
| 91H0 | 938.37 | 7.04 | 21.11 | 0.14 |
| 91I0 | 1 515.00 | 11.36 | 34.09 | 0.23 |
| 91J0 | 0.83 | 0.00 | 0.01 | 0.00 |
| 91K0 | 16 338.05 | 122.54 | 490.14 | 6.13 |
| 91L0 | 3 427.09 | 25.70 | 77.11 | 1.29 |
| 91M0 | 22 985.73 | 172.39 | 517.18 | 3.45 |
| 91P0 | 185.00 | 2.78 | 5.55 | 0.08 |
| 91Q0 | 20.18 | 0.15 | 0.30 | 0.00 |
| 91R0 | 85.49 | 0.64 | 1.28 | 0.02 |
| 91S0 | 268.02 | 2.01 | 8.04 | 0.10 |
| 91T0 | 276.78 | 0.00 | 4.15 | 0.06 |
| 91U0 | 9.16 | 0.00 | 0.14 | 0.00 |
| 91W0 | 2 123.92 | 15.93 | 63.72 | 0.80 |



Source: European Environment Agency – April 2021

| **EU27 excluding Romania** | Forest area ($km^2$) | Total Carbon Stock (Mt) | | Potential carbon sequestration rate (Mt $y^{-1}$) |
|---|---|---|---|---|
| | EU-27 | min | max | mean |
| 91Z0 | 389.14 | 2.92 | 8.76 | 0.06 |
| **Mediterranean forests** | **66 334.91** | **735.25** | **1572.82** | **13.99** |
| 9210 | 3271.37 | 24.54 | 98.14 | 1.23 |
| 9220 | 346.69 | 2.60 | 10.40 | 0.13 |
| 9230 | 13153.76 | 98.65 | 295.96 | 1.97 |
| 9240 | 3444.99 | 25.84 | 77.51 | 0.52 |
| 9250 | 457.31 | 3.43 | 10.29 | 0.07 |
| 9260 | 8527.79 | 63.96 | 255.83 | 1.92 |
| 9270 | 570.15 | 4.28 | 17.10 | 0.21 |
| 9280 | 996.00 | 7.47 | 29.88 | 0.37 |
| 9290 | 438.06 | 0.00 | 6.57 | 0.03 |
| 9310 | 3.47 | 0.03 | 0.08 | 0.00 |
| 9320 | 1966.46 | 14.75 | 29.50 | 0.15 |
| 9330 | 4283.86 | 64.26 | 96.39 | 0.96 |
| 9340 | 27784.22 | 416.76 | 625.15 | 6.25 |
| 9350 | 297.12 | 2.23 | 6.69 | 0.09 |
| 9360 | 602.94 | 4.52 | 9.04 | 0.05 |
| 9380 | 122.89 | 0.92 | 2.77 | 0.03 |
| 9390 | 61.40 | 0.92 | 1.38 | 0.01 |
| 93A0 | 6.44 | 0.10 | 0.14 | 0.00 |
| **Mountainous forests** | **50 947.05** | **255.36** | **950.57** | **9.56** |
| 9410 | 12 376.21 | 185.64 | 371.29 | 4.64 |
| 9420 | 4 006.57 | 0.00 | 60.10 | 0.90 |
| 9430 | 1 515.95 | 0.00 | 22.74 | 0.34 |
| 9510 | 82.78 | 1.24 | 1.86 | 0.02 |
| 9520 | 13.25 | 0.20 | 0.30 | 0.00 |
| 9530 | 7 896.00 | 59.22 | 118.44 | 1.78 |
| 9540 | 21 676.59 | 0.00 | 325.15 | 1.63 |
| 9550 | 736.98 | 5.53 | 11.05 | 0.06 |
| 9560 | 2 152.77 | 0.00 | 32.29 | 0.16 |
| 9570 | 0.95 | 0.00 | 0.01 | 0.00 |
| 9580 | 15.80 | 0.00 | 0.24 | 0.00 |
| 9590 | 2.91 | 0.00 | 0.04 | 0.00 |
| 95A0 | 470.29 | 3.53 | 7.05 | 0.04 |
| **TOTAL** | **357 951.65** | **2796.70** | **8816.44** | **100.97** |

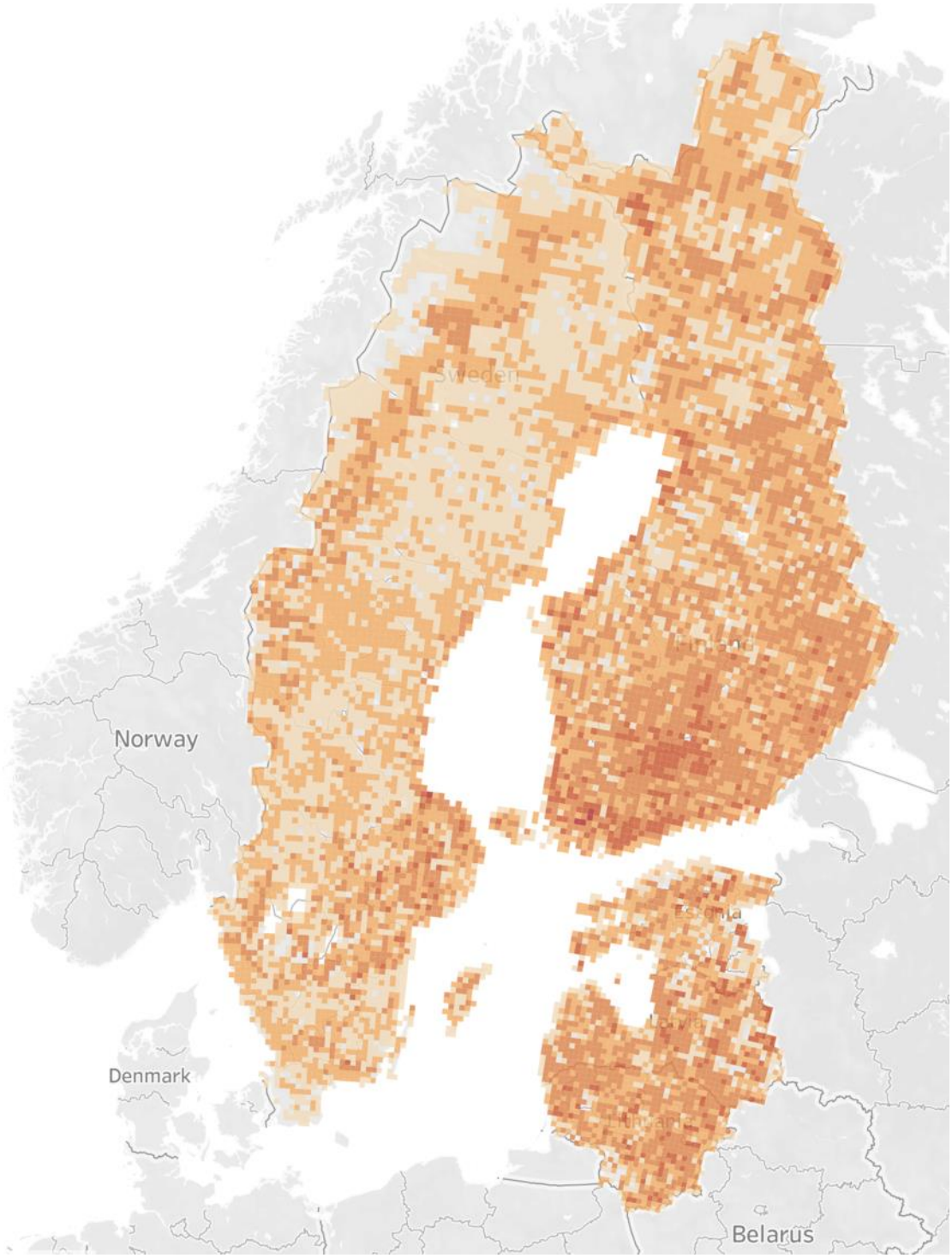
Note: areas reported by Romania note included



Source: European Environment Agency – April 2021

# Annex A

**Map 2 – Distribution of Annex I Boreal forest habitats (9010, 9020, 9030, 9040, 9050, 9060)**

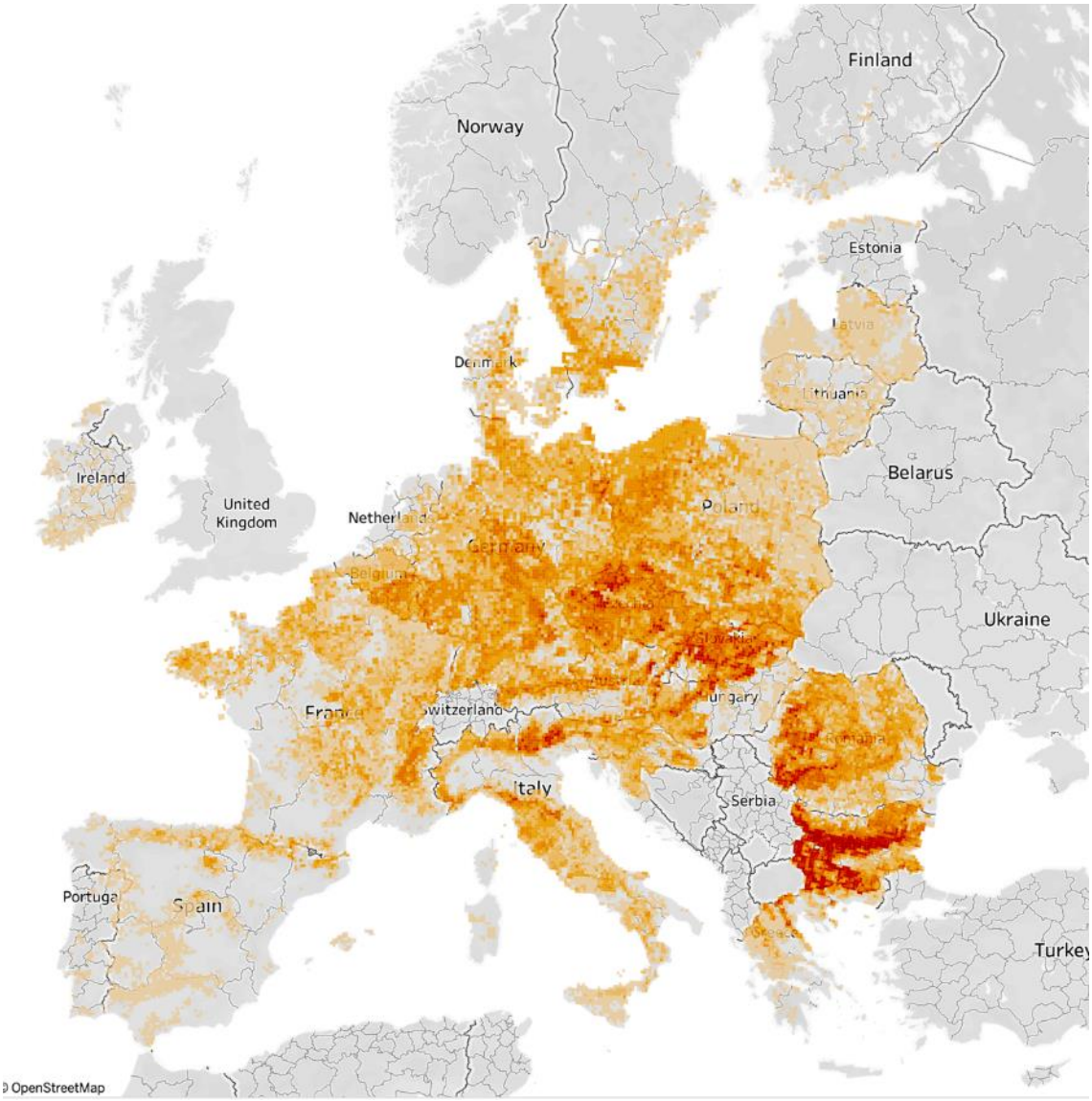



Note: the shades of brown indicate the number of habitat types per 10 km x 10 km grid cell.



Source: European Environment Agency – April 2021

**Map 3 – Distribution of Annex I Temperate forest habitats (9110-9150, 9170-91C0, 91G0-91M0, 91P0-91CA)**




Note: the shades of brown indicate the number of habitat types per 10 km x 10 km grid cell. Macaronesian islands not shown in the map.



Source: European Environment Agency – April 2021

**Map 4 – Distribution of Annex I Mediterranean forest habitats (9210-9290, 9310-9360, 9380-93A0)**

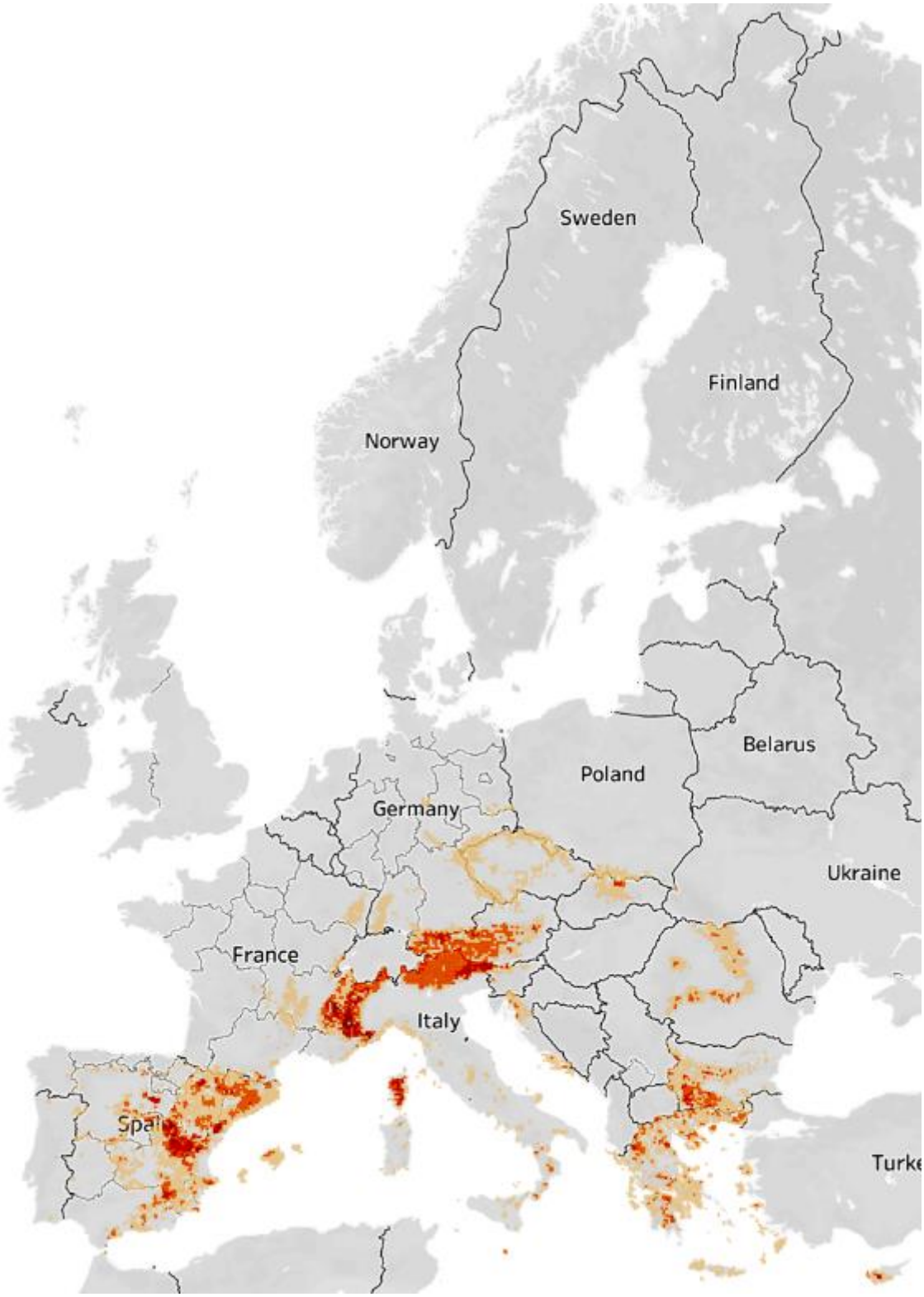



Note: the shades of brown indicate the number of habitat types per 10 km x 10 km grid cell. Macaronesian islands not shown in the map.



Source: European Environment Agency – April 2021

**Map 5 – Distribution of Annex I Mountainous forest habitats (9410-9430, 9510-95A0)**




Note: the shades of brown indicate the number of habitat types per 10 km x 10 km grid cell. Macaronesian islands not shown in the map.



Source: European Environment Agency – April 2021